

# Drone Garage Updates

2/5/2026

# To-Dos

- Get GPS coords and send to the rest of the mesh
- Fix re election stuff
- Fix the thing where nodes stop counting throughput
- Actually upload the code to the github this time

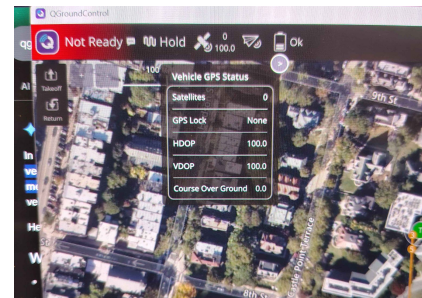
Had to redo our setup since a lot of the equipment in the lab was moved/taken out

# Getting GPS Coordinates

- Note: Not at the lab while writing, will try this stuff tmrw just spitballing for now
- So the connection is Drone  $\leftrightarrow$  Pi  $\leftrightarrow$  ESP
  - So have Pi have some script that uses MAVSDK in gathering information from the drone?
  - Afterwards, get that data and forward it to the ESP through the TX/RX thingies and in the ESP code have it broadcast (mark it by saying smth like “DRONE DATA: GPS: “ or have like some json thingy to keep things organized?)
- Another idea is to use the **mavlink-router**???
- Resources:
  - <https://discuss.px4.io/t/loading-gps-from-mavsdk/30433>
    - Forum for getting gps coordinates via: **drone.telemetry.gps\_info()** (using MAVSDK python stuff ofc)
  - <https://protonestiot.medium.com/communication-between-esp32-and-raspberry-pi-using-uart-7b776786c08a>
    - Skimmed thru this but looks possible (ESP and PI stuff)

# Getting GPS Coordinates

- On the Pi 5 setup
- In /Desktop/DroneGarage
  - New venv DGvenv with mavsdk and other packages
  - Script dg\_telem.py for connecting serially to the drone and esp for transferring telemetry data
    - Can connect to the drone successfully
    - Cant connect to the esp for some reason (serial.Serial() doesnt exist or smth like that???)
    - **GPS info is accessible but appears as like 0 and stuff like they dont exist**
      - **Solution is prob to take the drone setup outside**
        - Means we have to set up ssh stuff for the pi
      - **QGC when connected to the drone says the values (compare the images to the right) so the data extracted is correct we might just need to take it outside is my guess**
        - **Can get stuff like battery info so the data extraction works for sure**

[illegible][illegible]

# Getting GPS Coordinates

- Setting up Pi ssh (on Pi 5)
  - Sudo raspi-config > Interfacing Options > Enable (already done but to reproduce on Pi 4)
  - Hostname -l for ip
  - Whoami for name
  - Use like wsl or cygwin or whatever
- On wsl:
  - Note: This is on **stevens wifi** – just to get things up and running
  - Ssh pi@10.155.37.60 (at the time of writing)
    - Cant find its password so until then:
  - Probs do an ssh-keygen on your system
    - ssh-keygen -t rsa -b 4096
    - Get the key from id\_rsa.pub and put it into the Pi's .ssh/authorized\_keys then u can ssh and stuff

# Getting GPS Coordinates

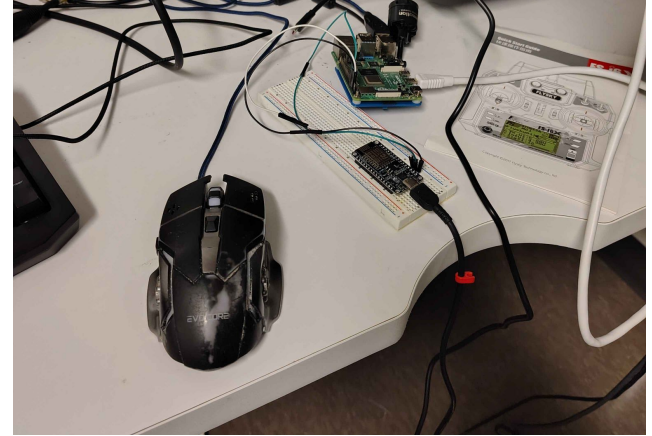
- For outdoor use, we will need portable power sources for the ESPs and the Pis
  - For now – prob lug around the drone setup(s) from the lab to that one rooftop in Babbio bc of the outlets there where we can have our setups plugged in and stuff
  -

# PI -> ESP -> PI

- Still need to figure out how to forward mavlink data throughout the rest of the mesh..
  - So pretty much just need to send the data from the pi -> esp over uart
  - Esp receives data sends it to its pi, where the pi will decode the information (looks like there are mavlink packages that will do this) and do whatever we need it to do
- We have already proven the method of sending the data from esp -> pi (see previous slides (11/14))
  - So now its just a matter of sending the new mavlink information (which we will do once we get the GPS coordinates)
- Also see (11/14) for setup
  - Need to configure all esp + pi setups (have 2 so far)

# Misc. – New Setups

- One of the monitors is bricked so use that tv screen (HDMI on the left side)
- Connected rn to it is the Pi 4 with mavlink-router already set up along with a venv in a folder labeled “Drone Garage” with mavsdk and other packages installed



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS bash + - - - - -  
(Dovev) aeroshield@aeroshield:~/DroneGarage $ pip install RPi.GPIO  
Getting requirements to build wheel ... done  
Preparing metadata (pyproject.toml) ... done  
Building wheels for collected packages: RPi.GPIO  
Building wheel for RPi.GPIO (pyproject.toml) ... done  
Created wheel for RPi.GPIO: filename=rpi_gpio-0.7.1-cp313-cp313-linux_aarch64.whl  
size=69284 sha256=2d98a8ab7c13df1a417e98dc5fd4650f5ec719232a6d11e574237520c618065d  
Stored in directory: /home/aeroshield/.cache/pip/wheels/f1/71/0b/b4cf15eed2218094  
44483ab72446deeb203f4aa2c185ec1c  
Successfully built RPi.GPIO  
Installing collected packages: RPi.GPIO  
Successfully installed RPi.GPIO-0.7.1  
(Dovev) aeroshield@aeroshield:~/DroneGarage $
```